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| <b>SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE</b>                                      |                                                                                                           | <b>DEPARTMENT OF COMPUTER SCIENCE ENGINEERING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Program Name:</b> B. Tech                                                                       |                                                                                                           | <b>Assignment Type:</b> Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Academic Year:</b> 2025-2026 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Course Coordinator Name</b>                                                                     |                                                                                                           | Dr. Rishabh Mittal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Instructor(s) Name</b>                                                                          |                                                                                                           | <table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table> |                                 | Mr. S Naresh Kumar | Ms. B. Swathi | Dr. Sasanko Shekhar Gantayat | Mr. Md Sallauddin | Dr. Mathivanan | Mr. Y Srikanth | Ms. N Shilpa | Dr. Rishabh Mittal (Coordinator) | Dr. R. Prashant Kumar | Mr. Ankushavali MD | Mr. B Viswanath | Ms. Sujitha Reddy | Ms. A. Anitha | Ms. M.Madhuri | Ms. Katherashala Swetha | Ms. Velpula sumalatha | Mr. Bingi Raju |
| Mr. S Naresh Kumar                                                                                 |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. B. Swathi                                                                                      |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Sasanko Shekhar Gantayat                                                                       |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Md Sallauddin                                                                                  |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Mathivanan                                                                                     |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Y Srikanth                                                                                     |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. N Shilpa                                                                                       |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. Rishabh Mittal (Coordinator)                                                                   |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Dr. R. Prashant Kumar                                                                              |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Ankushavali MD                                                                                 |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. B Viswanath                                                                                    |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Sujitha Reddy                                                                                  |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. A. Anitha                                                                                      |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. M.Madhuri                                                                                      |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Katherashala Swetha                                                                            |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Ms. Velpula sumalatha                                                                              |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| Mr. Bingi Raju                                                                                     |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>CourseCode</b>                                                                                  | 23CS002PC304                                                                                              | <b>Course Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AI Assisted Coding              |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Year/Sem</b>                                                                                    | III/II                                                                                                    | <b>Regulation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R23                             |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Date and Day of Assignment</b>                                                                  | Week4 – Monday                                                                                            | <b>Time(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 23CSBTB01 To 23CSBTB52          |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Duration</b>                                                                                    | 2 Hours                                                                                                   | <b>Applicable to Batches</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | All batches                     |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Assignment Number: 8.1</b> (Present assignment number)/ <b>24</b> (Total number of assignments) |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
|                                                                                                    |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| <b>Q.No.</b>                                                                                       | <b>Question</b>                                                                                           | <b>Expected Time to complete</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |
| 1                                                                                                  | Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases<br><b>Lab Objectives:</b> | Week4 - Monday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |                    |               |                              |                   |                |                |              |                                  |                       |                    |                 |                   |               |               |                         |                       |                |

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|  | <ul style="list-style-type: none"> <li>• To introduce students to test-driven development (TDD) using AI code generation tools.</li> <li>• To enable the generation of test cases before writing code implementations.</li> <li>• To reinforce the importance of testing, validation, and error handling.</li> <li>• To encourage writing clean and reliable code based on AI-generated test expectations.</li> </ul> <p><b>Lab Outcomes (LOs):</b></p> <p>After completing this lab, students will be able to:</p> <ul style="list-style-type: none"> <li>• Use AI tools to write test cases for Python functions and classes.</li> <li>• Implement functions based on test cases in a test-first development style.</li> <li>• Use unittest or pytest to validate code correctness.</li> <li>• Analyze the completeness and coverage of AI-generated tests.</li> <li>• Compare AI-generated and manually written test cases for quality and logic</li> </ul> <p>Task Description #1 (Password Strength Validator – Apply AI in Security Context)</p> <ul style="list-style-type: none"> <li>• Task: Apply AI to generate at least 3 assert test cases for <code>is_strong_password(password)</code> and implement the validator function.</li> <li>• Requirements: <ul style="list-style-type: none"> <li>○ Password must have at least 8 characters.</li> <li>○ Must include uppercase, lowercase, digit, and special character.</li> <li>○ Must not contain spaces.</li> </ul> </li> </ul> <p>Example Assert Test Cases:</p> <pre>assert is_strong_password("Abcd@123") == True assert is_strong_password("abcd123") == False assert is_strong_password("ABCD@1234") == True</pre> <p>Expected Output #1:</p> <ul style="list-style-type: none"> <li>• Password validation logic passing all AI-generated test cases</li> <li>•</li> </ul> <pre>. import re  def is_strong_password(password):</pre> |  |
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|  | <pre> if len(password) &lt; 8 or " " in password:     return False  has_upper = any(c.isupper() for c in password) has_lower = any(c.islower() for c in password) has_digit = any(c.isdigit() for c in password) has_special = bool(re.search(r"[!@#\$%^&amp;*(),.\?\"':{} &lt;&gt;]", password))  return all([has_upper, has_lower, has_digit, has_special])  # Assert Test Cases assert is_strong_password("Abcd@123") == True assert is_strong_password("abcd123") == False assert is_strong_password("ABCD@1234") == False # Missing lowercase assert is_strong_password("Abc @123") == False # Contains space #----- Task Description #2 (Number Classification with Loops – Apply AI for Edge Case Handling) <ul style="list-style-type: none"> <li>Task: Use AI to generate at least 3 assert test cases for a classify_number(n) function. Implement using loops.</li> <li>Requirements: <ul style="list-style-type: none"> <li>Classify numbers as Positive, Negative, or Zero.</li> <li>Handle invalid inputs like strings and None.</li> <li>Include boundary conditions (-1, 0, 1).</li> </ul> </li> </ul> Example Assert Test Cases: assert classify_number(10) == "Positive" assert classify_number(-5) == "Negative" assert classify_number(0) == "Zero" Expected Output #2: <ul style="list-style-type: none"> <li>Classification logic passing all assert tests.</li> </ul>  def classify_number(n):     if not isinstance(n, (int, float)):         return "Invalid Input"      if n &gt; 0:         return "Positive" </pre> |  |
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```
elif n < 0:
    return "Negative"
else:
    return "Zero"
```

```
# Assert Test Cases
assert classify_number(10) == "Positive"
assert classify_number(-5) == "Negative"
assert classify_number(0) == "Zero"
assert classify_number("abc") == "Invalid Input"
#-----
```

### Task Description #3 (Anagram Checker – Apply AI for String Analysis)

- Task: Use AI to generate at least 3 assert test cases for `is_anagram(str1, str2)` and implement the function.
- Requirements:
  - Ignore case, spaces, and punctuation.
  - Handle edge cases (empty strings, identical words).

Example Assert Test Cases:

```
assert is_anagram("listen", "silent") == True
assert is_anagram("hello", "world") == False
assert is_anagram("Dormitory", "Dirty Room") == True
```

Expected Output #3:

- Function correctly identifying anagrams and passing all AI-generated tests.

```
import string
```

```
def is_anagram(str1, str2):
    def clean_str(s):
        s = s.lower().replace(" ", "")
        return sorted([char for char in s if char not in
string.punctuation])
```

```
    return clean_str(str1) == clean_str(str2)
```

```
# Assert Test Cases
assert is_anagram("listen", "silent") == True
assert is_anagram("hello", "world") == False
```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
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|  | <pre> assert is_anagram("Dormitory", "Dirty Room") == True  #----- Task Description #4 (Inventory Class – Apply AI to Simulate Real-World Inventory System)     • Task: Ask AI to generate at least 3 assert-based tests for an Inventory class with stock management.     • Methods:         ◦ add_item(name, quantity)         ◦ remove_item(name, quantity)         ◦ get_stock(name) Example Assert Test Cases: inv = Inventory() inv.add_item("Pen", 10) assert inv.get_stock("Pen") == 10 inv.remove_item("Pen", 5) assert inv.get_stock("Pen") == 5 inv.add_item("Book", 3) assert inv.get_stock("Book") == 3 Expected Output #4:     • Fully functional class passing all assertions. class Inventory:     def __init__(self):         self.stock = {}      def add_item(self, name, quantity):         self.stock[name] = self.stock.get(name, 0) + quantity      def remove_item(self, name, quantity):         if name in self.stock and self.stock[name] &gt;= quantity:             self.stock[name] -= quantity         else:             return "Insufficient stock"      def get_stock(self, name):         return self.stock.get(name, 0)  # Assert Test Cases inv = Inventory() inv.add_item("Pen", 10) </pre> |  |
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|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>assert inv.get_stock("Pen") == 10 inv.remove_item("Pen", 5) assert inv.get_stock("Pen") == 5 inv.add_item("Book", 3) assert inv.get_stock("Book") == 3</pre> <p>#-----</p> <p>Task Description #5 (Date Validation &amp; Formatting – Apply AI for Data Validation)</p> <ul style="list-style-type: none"><li>• Task: Use AI to generate at least 3 assert test cases for <code>validate_and_format_date(date_str)</code> to check and convert dates.</li><li>• Requirements:<ul style="list-style-type: none"><li>◦ Validate "MM/DD/YYYY" format.</li><li>◦ Handle invalid dates.</li><li>◦ Convert valid dates to "YYYY-MM-DD".</li></ul></li></ul> <p>Example Assert Test Cases:</p> <pre>assert validate_and_format_date("10/15/2023") == "2023-10-15" assert validate_and_format_date("02/30/2023") == "Invalid Date" assert validate_and_format_date("01/01/2024") == "2024-01-01"</pre> <p>Expected Output #5:</p> <ul style="list-style-type: none"><li>• Function passes all AI-generated assertions and handles edge cases.</li></ul> <pre>from datetime import datetime  def validate_and_format_date(date_str):     try:         date_obj = datetime.strptime(date_str, "%m/%d/%Y")         return date_obj.strftime("%Y-%m-%d")     except ValueError:         return "Invalid Date"</pre> <p># Assert Test Cases</p> <pre>assert validate_and_format_date("10/15/2023") == "2023-10-15" assert validate_and_format_date("02/30/2023") == "Invalid Date" assert validate_and_format_date("01/01/2024") == "2024-01-01"</pre> |  |
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